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Semiconductor packages

Abstract

A semiconductor package includes a first integrated circuit, a plurality of first through vias and a plurality of fin-shaped through vias. The first through vias surround the first integrated circuit. The fin-shaped through vias are physically connected to the first through vias respectively, wherein the first through vias are disposed between the first integrated circuit and the fin-shaped through vias.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8993380	12/2014	Hou et al.	N/A	N/A
9281254	12/2015	Yu et al.	N/A	N/A
9299649	12/2015	Chiu et al.	N/A	N/A
9372206	12/2015	Wu et al.	N/A	N/A
9425126	12/2015	Kuo et al.	N/A	N/A
9443783	12/2015	Lin et al.	N/A	N/A
9461018	12/2015	Tsai et al.	N/A	N/A
9496189	12/2015	Yu et al.	N/A	N/A
9666502	12/2016	Chen et al.	N/A	N/A
9735131	12/2016	Su et al.	N/A	N/A
2020/0006196	12/2019	Lin	N/A	H01L 23/5389
2022/0122936	12/2021	Williamson	N/A	H01L 24/73

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Background/Summary

BACKGROUND

(1) The integrated circuit (IC) industry has experienced a fast-paced growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component or line that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1A to FIG. 1G illustrate schematic cross-sectional views of a method of forming a semiconductor package in accordance with some embodiments of the disclosure.

(3) FIG. 2A illustrates a schematic top view of openings for forming through vias and thermal conductive patterns of the semiconductor package in accordance with some embodiments of the disclosure, and FIG. 2B illustrates a schematic top view of the through vias and the thermal conductive patterns of the semiconductor package in accordance with some embodiments of the disclosure.

(4) FIG. 3 illustrates a schematic cross-sectional view of a semiconductor package in accordance with some embodiments of the disclosure.

(5) FIG. 4A and FIG. 4B respectively illustrate a schematic cross-sectional view and a schematic top view of a semiconductor package in accordance with some embodiments.

(6) FIG. 5 illustrates a schematic top view of a semiconductor package in accordance with some embodiments of the disclosure.

(7) FIG. 6 illustrates a schematic top view of a semiconductor package in accordance with some embodiments of the disclosure.

(8) FIG. 7 illustrates a schematic top view of a semiconductor package in accordance with some embodiments of the disclosure.

DETAILED DESCRIPTION

(9) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(10) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(11) Other features and processes may also be included. For example, testing structures may be included to aid in the verification testing of the 3D packaging or 3DIC devices. The testing structures may include, for example, test pads formed in a redistribution layer or on a substrate that allows the testing of the 3D packaging or 3DIC, the use of probes and/or probe cards, and the like. The verification testing may be performed on intermediate structures as well as the final structure. Additionally, the structures and methods disclosed herein may be used in conjunction with testing

methodologies that incorporate intermediate verification of known good integrated circuits to increase the yield and decrease costs.

(12) FIG. 1A to FIG. 1G illustrate schematic cross-sectional views of a method of forming a semiconductor package in accordance with some embodiments of the disclosure. FIG. 2A illustrates a schematic top view of openings for forming through vias and thermal conductive patterns of the semiconductor package in accordance with some embodiments of the disclosure, and FIG. 2B illustrates a schematic top view of the through vias and the thermal conductive patterns of the semiconductor package in accordance with some embodiments of the disclosure. FIG. 1C is a schematic cross-sectional view along line I-I' of FIG. 2A, and FIG. 1E is a schematic cross-sectional view along line I-I' of FIG. 2B, for example. It is understood that the disclosure is not limited by the method described below. Additional operations can be provided before, during, and/or after the method and some of the operations described below can be replaced or eliminated, for additional embodiments of the methods.

(13) Although FIG. 1A to FIG. 1G are described in relation to a method, it is appreciated that the structures disclosed in FIG. 1A to FIG. 1G are not limited to such a method, but instead may stand alone as structures independent of the method.

(14) Referring to FIG. 1A, a semiconductor structure **101** is provided on a carrier **102**. In some embodiments, the carrier **102** is a supporting substrate. For example, the carrier **102** is a glass, a semiconductor wafer or a semiconductor block. The carrier **102** may be made of a semiconductor material such as silicon, germanium, diamond, or the like. Alternatively, compound materials such as silicon germanium, silicon carbide, gallium arsenic, indium arsenide, indium phosphide, silicon germanium carbide, gallium arsenic phosphide, gallium indium phosphide, combinations thereof, and the like, may also be used. In some embodiments, the carrier **102** is un-doped and/or free of active devices. However, the disclosure is not limited thereto.

(15) In some embodiments, an adhesive layer **104** is formed on the carrier **102**, and then a semiconductor structure **101** is picked and placed on the adhesive layer **104** over the carrier **102** through the adhesive layer **104**. The adhesive layer **104** may be a die attach film (DAF) or other suitable adhesive material. The adhesive layer **104** is formed on an entire surface of the carrier **102**, for example. In other words, the adhesive layer **104** is continuously extended between the carrier **102** and the semiconductor structure **101**. In such embodiments, a sidewall of the adhesive layer **104** is substantially flush with a sidewall of the carrier **102**. However, the disclosure is not limited thereto. In alternative embodiments (not shown), a sidewall of the adhesive layer **104** is substantially flush with a sidewall of the semiconductor structure **101**. In some embodiments, a debonding layer (not shown) is disposed between the adhesive layer **104** and the carrier **102**, and the material of the debonding layer is suitable for bonding and debonding the carrier **102** from the above layers or die disposed thereon. In some embodiments, the debonding layer includes a light-to-heat conversion ("LTHC") layer, and such layer enables debonding from the carrier by applying laser irradiation. In some embodiments, a dielectric layer is further disposed between the debonding layer and the carrier **102** or replaces with the adhesive layer **104**, and the dielectric layer is made of a dielectric material including silicon oxide, benzocyclobutene ("BCB"), polybenzooxazole ("PBO") or other suitable polymer-based dielectric material. In some embodiments, the carrier **102** has scribe lines SL to define a region for an integrated circuit **110** including the semiconductor structure **101**. For example, the integrated circuit **110** is disposed between adjacent scribe lines SL. For clarity, only one integrated circuit **110** (e.g., also only one semiconductor structure **101**) is illustrated. However, a plurality of integrated circuits **110** (e.g., also a plurality of semiconductor structures **101**) may be respectively provided on regions of the carrier **102** defined by the scribe lines SL. In such embodiments, the integrated circuits **110** (e.g., also the semiconductor structures **101**) are arranged side by side. For example, the integrated circuits **110** are arranged in a matrix.

(16) In some embodiments, the semiconductor structure **101** includes a semiconductor substrate

112, an active device layer **113** on the semiconductor substrate **112** and a plurality of through vias **115** in a dielectric layer **117**. The semiconductor substrate **112** may be made of a semiconductor material such as silicon, germanium, diamond, or the like. Alternatively, compound materials such as silicon germanium, silicon carbide, gallium arsenic, indium arsenide, indium phosphide, silicon germanium carbide, gallium arsenic phosphide, gallium indium phosphide, combinations thereof, and the like, may also be used. The active device layer **113** may be partially in the semiconductor substrate **112** or in a dielectric layer on the semiconductor substrate **112**. The active device layer **113** includes a dielectric layer and a plurality of active devices and a plurality of conductive patterns in the dielectric layer, for example. However, the disclosure is not limited thereto. In alternative embodiments, the active device layer **113** may have other configurations.

(17) Then, an encapsulant **120** is formed over the carrier **102** to encapsulate the semiconductor structure **101**. In some embodiments, the encapsulant **120** is formed by forming an insulating material to cover the semiconductor structure **101** and fill up gaps (e.g., gaps between the semiconductor structures **101**), and then performing a planarization process on the insulating material until the semiconductor structure **101** and the insulating material are accessibly revealed. The insulating material may include a dielectric material such as silicon oxide, TEOS silicon oxide, or other suitable gap-filling materials formed by a deposition process such as CVD, PECVD, or other suitable method. Formation of the insulating material may further include a gap-filling process, a curing process and/or a hardening process. The planarization process may include a grinding process, a chemical mechanical polishing (CMP) process, or other suitable removing process. In some embodiments, a surface (e.g., top surface) of the encapsulant **120** is substantially coplanar with a surface (e.g., top surface) of the semiconductor structure **101**.

(18) Then, a bonding layer **114** and a plurality of bonding pads **116** in the bonding layer **114** are formed over the encapsulant **120** and the semiconductor structure **101**, for example. In some embodiments, the bonding layer **114** includes organic materials such as polybenzoxazole (PBO) and polyimide (PI) or inorganic materials such as silicon oxide or silicon nitride. The bonding layer **114** may be a single layer or a multiple layer structure. The bonding pads **116** may be vias, pads, pillars or other suitable connectors. In some embodiments, the bonding pads **116** are electrically connected to the through vias **115**. A material of the bonding pads **116** includes copper (Cu), aluminum (Al), aluminum copper alloy (AlCu), gold, titanium, cobalt, an alloy, or other suitable conductive material, for example. Then, the integrated circuit **110** is formed. In some embodiments, the integrated circuit **110** is a wafer or a die. For example, the integrated circuit **110** is a logic die or a system-on-chip (SoC).

(19) After that, a plurality of integrated circuits **130** are bonded to the integrated circuit **110** over the carrier **102**. The integrated circuit **130** may be different from the integrated circuit **110** in size and/or type. For example, the integrated circuit **130** is smaller than the integrated circuit **110**, and multiple integrated circuits **130** are arranged over the same integrated circuit **110**. The integrated circuits **130** may be arranged in a matrix. The integrated circuit **130** may include a die. The integrated circuit **130** is a single die, a chiplet or a die stack, for example. In some embodiments, the integrated circuit **110** is a logic die, and the integrated circuit **130** is a memory die. For example, the integrated circuit **130** is a DRAM die, a SRAM die, a flash integrated circuit, a high bandwidth memory (HBM) die, an application-specific integrated circuit (ASIC) die, a central processing unit (CPU) die, a graphic processing unit (GPU) die, a micro control unit (MCU) die, a Transceiver (TRX) die, an input-output (I/O) die, a baseband (BB) die or an application processor (AP) die. However, the disclosure is not limited thereto. The integrated circuit **110** and the integrated circuits **130** have the same size and/or type according to the requirements.

(20) The integrated circuit **130** may include a semiconductor substrate **132**, a bonding layer **134** on the semiconductor substrate **132** and a plurality of bonding pads **136** in the bonding layer **134**. The semiconductor substrate **132** may be made of a semiconductor material such as silicon, germanium, diamond, or the like. Alternatively, compound materials such as silicon germanium, silicon carbide,

gallium arsenic, indium arsenide, indium phosphide, silicon germanium carbide, gallium arsenic phosphide, gallium indium phosphide, combinations thereof, and the like, may also be used. In some embodiments, the bonding layer **134** includes organic materials such as polybenzoxazole (PBO) and polyimide (PI) or inorganic materials. The bonding pads **136** may be vias, pads, pillars or other suitable connectors. A material of the bonding pads **136** includes copper (Cu), aluminum (Al), aluminum copper alloy (AlCu), gold, titanium, cobalt, an alloy, or other suitable conductive material, for example. In some embodiments, a thickness of the integrated circuit **130** (e.g., about a thickness of the semiconductor substrate **132**) is in a range of 80 μm to 160 μm . A spacing between the integrated circuits **130** (i.e., die to die distance) is in a range of 60 μm to 100 μm , and a distance between the integrated circuit **130** and the scribe line SL of the carrier **102** is in a range of 3 mm to 5 mm.

(21) In some embodiments, the bonding pads **136** of the integrated circuit **130** are bonded to the bonding pads **116** of the integrated circuit **110**, and the bonding layer **134** of the integrated circuit **130** is bonded to the bonding layer **114** of the integrated circuit **110**. Thus, the integrated circuits **130** are electrically connected to the integrated circuit **110** respectively.

(22) Referring to FIG. **1B**, a seed layer **140** is formed on the integrated circuits **130** over the integrated circuit **110**. In some embodiments, the seed layer **140** is continuously formed on exposed surfaces of the encapsulant **120**, the integrated circuit **110** and the integrated circuits **130**. For example, the seed layer **140** is continuously formed on a top surface of the encapsulant **120**, a top surface of the integrated circuit **110**, and sidewalls and top surfaces of the integrated circuit **110**. In some embodiments, the seed layer **140** includes titanium (Ti)/copper (Cu) or other suitable conductive material, and the seed layer **140** is formed by electrochemical plating (ECP), electroplating, electroless plating or other suitable process. In alternative embodiments, the seed layer **140** is also referred to as a barrier layer.

(23) Referring to FIG. **1C**, a resist layer **150** is formed over the carrier **102** to cover the integrated circuits **130**, and the resist layer **150** has a plurality of openings **152a**, **152b** and **154** therein. The resist layer **150** is a dry film resist (DFR) layer, for example. In some embodiments, the openings **152a**, **152b** and **154** expose portions of the seed layer **140** on the encapsulant **120**. In some embodiments, the resist layer **150** is formed on the seed layer **140** to fully cover the integrated circuits **130**. The resist layer **150** may be conformally formed on the encapsulant **120** and the integrated circuits **130**. Thus, as shown in FIG. **1C**, a surface (e.g., top surface) of the resist layer **150** over the integrated circuits **130** may be higher than a surface (e.g., top surface) of the resist layer **150** over the integrated circuit **110** and the encapsulant **120**. In some embodiments, the resist layer **150** has a first portion and a second portion on the first portion, and the first portion is rectangular and the second portion is trapezoid.

(24) In some embodiments, as shown in FIG. **2A**, the openings **152a**, **152b** and **154** are formed to surround the integrated circuits **130**, respectively. For example, the openings **152a**, **152b** and **154** are formed along one ring-shaped path surrounding the integrated circuits **130**, respectively. In some embodiments, the opening **152a** is separated from the opening **152b** and disposed between the integrated circuits **130** and the opening **152b**. The opening **154** is connected to the opening **152b**, in other words, the opening **154** is integrally formed with the opening **152b**, for example. In some embodiments, as shown in FIG. **2A**, from a top view, a shape of the opening **152b** is substantially the same as the opening **152a**, and a shape of the opening **154** is different from the opening **152a**, **152b**. For example, the opening **152a** and the opening **152b** are circular, and the opening **154** is line-shaped (e.g., fin-shaped). In such embodiments, the openings **152a** and the openings **152b** are formed by using the same photomask, and the openings **154** are formed by using another photomask. However, the disclosure is not limited thereto. The openings **152a**, **152b** and **154** may be formed by using the same photomask. A diameter (e.g., width and/or length) of the opening **154** may be substantially the same as, smaller than or larger than the opening **152a**, **152b**. In some embodiments, a width of the opening **154** is smaller than the opening **152a**, **152b**, and a

length of the opening **154** is larger than the opening **152a**, **152b**. In alternative embodiments, the opening **152b** and the opening **154** are collectively referred to as an elongated opening, and the opening **152b** and the opening **154** are respectively a via-shaped portion and a fin-shaped portion of the elongated opening.

(25) Referring to FIG. 1D, a plurality of through vias **160a**, **160b** are formed in the openings **152a** and **152b**, and a plurality of thermal conductive patterns **162** are formed in the openings **154**. In some embodiments, the through vias **160a**, **160b** and the thermal conductive patterns **162** are formed to have a height smaller than a height of the resist layer **150**. In some embodiments, the height of the through vias **160a**, **160b** and the thermal conductive patterns **162** is also smaller than the height of the integrated circuits **130**. In such embodiments, surfaces (e.g., top surfaces) of the through vias **160a**, **160b** and the thermal conductive patterns **162** are lower than surfaces (e.g., top surfaces) of the resist layer **150** and the integrated circuits **130**. The through vias **160a**, **160b** and the thermal conductive patterns **162** may have substantially the same height, and thus the surfaces (e.g., top surfaces) of the through vias **160a**, **160b** and the thermal conductive patterns **162** are substantially coplanar. The height of the through vias **160a**, **160b** and the thermal conductive patterns **162** is in a range of 50 μm to 300 μm , for example, depending on the height of the integrated circuits **130**. In some embodiments, the through vias **160a**, **160b** and the thermal conductive patterns **162** are formed by forming a conductive material in the openings **152a**, **152b** and **154** without filling up the openings **152a**, **152b** and **154**. The conductive material may include copper (Cu), aluminum (Al), aluminum copper alloy (AlCu), gold, titanium, cobalt, an alloy, or other suitable conductive material. The conductive material may be formed by electrochemical plating (ECP), electroplating, electroless plating or other suitable process.

(26) Referring to FIG. 1E, the resist layer **150** is removed. The resist layer **150** may be removed by a stripping process. In some embodiments, the through vias **160a**, **160b** and the thermal conductive patterns **162** are formed on the seed layer **140** directly. In some embodiments, the through vias **160a**, **160b** and the thermal conductive patterns **162** are not physically connected to the integrated circuits **110**, **130**. However, the disclosure is not limited thereto. In alternative embodiments, at least one of the through vias **160a**, **160b** and the thermal conductive pattern **162** is physically connected to one of the integrated circuits **110**, **130**. For example, at least one of the through vias **160a**, **160b** and the thermal conductive pattern **162** is physically connected to the integrated circuit **110**, **130** through the bonding pad **116**, **136**.

(27) In some embodiments, as shown in FIG. 2B, the through vias **160a**, **160b** and the thermal conductive patterns **162** are formed to surround the integrated circuits **130**, respectively. For example, the through vias **160a**, **160b** and the thermal conductive patterns **162** are formed along one ring-shaped path surrounding the integrated circuits **130**, respectively. In some embodiments, the through vias **160a** are arranged along two ring-shaped paths surrounding the integrated circuits **130**. However, the disclosure is not limited thereto. In alternative embodiments, the through vias **160a** are arranged along one or more than two ring-shaped paths surrounding the integrated circuits **130**.

(28) In some embodiments, the through via **160a** is separated from the through via **160b** and disposed between the integrated circuits **130** and the through via **160b**. In some embodiments, the thermal conductive pattern **162** is physically connected to the through via **160b**, in other words, the thermal conductive pattern **162** is integrally formed with the through via **160b**. Thus, no interface exists between the thermal conductive pattern **162** and the through via **160b**, for example. As shown in FIG. 2B, from a top view, a shape of the through via **160b** is substantially the same as the through via **160a**, and a shape of the thermal conductive pattern **162** is different from the through via **160a**, **160b**. For example, the through via **160a**, **160b** are circular, and the thermal conductive pattern **162** is line-shaped (e.g., fin-shaped). However, the disclosure is not limited thereto. In alternative embodiments, the thermal conductive pattern **162** may have other suitable shapes. A diameter (e.g., width and/or length) of the thermal conductive pattern **162** may be substantially the

same as, smaller than or larger than that of the through via **160a**, **160b**. For example, the width of the thermal conductive pattern **162** is smaller than the through via **160a**, **160b**. The width of the through via **160a**, **160b** and the thermal conductive patterns **162** may be in a range of 10 μm to 500 μm . The length of the thermal conductive pattern **162** may be larger than the through via **160a**, **160b**. The length of the thermal conductive pattern **162** may be in a range of 10 μm to 500 μm . In an embodiment in which the thermal conductive pattern **162** may have a width substantially the same as the through via **160b**, the through via **160b** is integrally formed with the thermal conductive pattern **162** as a wall structure having a constant width. In some embodiments, the thermal conductive pattern **162** is physically connected to the through via **160b** and extends from a sidewall of the through via **160b** toward a sidewall of the encapsulant **120**. For example, the thermal conductive pattern **162** extends on the integrated circuit **110** and the encapsulant **120** from the through via **160b** to the scribe line SL of the carrier **102**. In some embodiments, the thermal conductive pattern **162** extends beyond the scribe line SL of the carrier **102**. In alternative embodiments, a sidewall of the thermal conductive pattern **162** is substantially flush with the scribe line SL of the carrier **102**. The through via **160b** and the thermal conductive pattern **162** may be collectively referred to as a dummy through via, and the through via **160b** and the thermal conductive pattern **162** are, for example, respectively a via-shaped portion and an extension portion (e.g., fin-shaped portion) of the dummy through via.

(29) In some embodiments, as shown in FIG. 2B, the thermal conductive patterns **162** at the same side of the integrated circuit **110** are substantially parallel to one another. The thermal conductive patterns **162** may be arranged regularly, that is, a spacing between the adjacent thermal conductive patterns **162** is constant. In alternative embodiments, the thermal conductive patterns **162** at the same side of the integrated circuit **110** may be arranged irregularly or randomly, that is, a spacing between the adjacent thermal conductive patterns **162** is not constant. Similarly, the through vias **160a** may be arranged regularly, irregularly or randomly. In some embodiments, the through vias **160a** are also referred to as internal through vias, and the through vias **160b** are also referred to as external through vias.

(30) After removing the resist layer **150**, an encapsulant **170** is formed to encapsulate the integrated circuits **130** and the through vias **160a**. In some embodiments, the encapsulant **170** is formed by forming an insulating material to cover the integrated circuits **130** and fill up gaps among the integrated circuits **130**, the through vias **160a** and the through via **160b**, and then performing a planarization process on the insulating material until the integrated circuits **130** and the insulating material are accessibly revealed. The insulating material may include a dielectric material such as silicon oxide, TEOS silicon oxide, or other suitable gap-filling materials formed by a deposition process such as CVD, PECVD, or other suitable method. Formation of the insulating material may further include a gap-filling process, a curing process and/or a hardening process. The planarization process may include a grinding process, a chemical mechanical polishing (CMP) process, or other suitable removing process. In some embodiments, after forming the encapsulant **170**, a surface (e.g., top surface) of the encapsulant **170** is substantially coplanar with surfaces (e.g., top surfaces) of the integrated circuits **130**. For example, the surface (e.g., top surface) of the encapsulant **170** is substantially coplanar with a surface (e.g., top surfaces) of the seed layer **140** on the integrated circuits **130**. In some embodiments, the seed layer **140** on the integrated circuits **130** may be partially or entirely removed during the planarization process. In other words, after the planarization process, the seed layer **140** may be formed on the sidewalls of the integrated circuits **130** without covering surfaces (e.g., top surfaces) of the integrated circuits **130**. The surface (e.g., top surface) of the encapsulant **170** may be substantially coplanar with surfaces (e.g., top surfaces) of the seed layer **140** and the integrated circuits **130**.

(31) In some embodiments, since the surfaces (e.g., top surfaces) of the through vias **160a**, the through via **160b** and the thermal conductive pattern **162** are lower than the surface (e.g., top surfaces) of the integrated circuits **130**, during the planarization process, the through vias **160a**, the

through via **160b** and the thermal conductive pattern **162** would not be planarized. Accordingly, contaminant (e.g., copper residue) generated by performing the planarization process on materials (e.g., copper) of the through vias **160a**, the through via **160b** and the thermal conductive pattern **162** in another embodiment may be prevented. In some embodiments, after planarization, portions of the insulating material are removed to form recesses **172**, and the recesses **172** expose portions of through vias **160a** and portions of the through via **160b** and the thermal conductive pattern **162**. For example, the through vias **160a** and the through via **160b** are exposed by the recesses **172**. The recess **172** may have a depth in a range of 2 μm to 5 μm . In alternative embodiments, at least one of the recesses **172** is omitted. For example, the encapsulant **170** is formed without the recesses **172**.

(32) Then, a passivation layer **174** may be formed on the encapsulant **170**, to protect the integrated circuits **130**. In some embodiments, the passivation layer **174** includes organic materials such as polybenzoxazole (PBO) and polyimide (PI) or inorganic materials. In some embodiments, the recesses **172** are filled up by the passivation layer **174**. However, the disclosure is not limited thereto. In alternative embodiments, the recesses **172** may be partially or entirely retained. In addition, the passivation layer **174** may be omitted.

(33) Referring to FIG. 1F, a dicing (cutting) process is performed, so as to form a semiconductor package **100**. In some embodiments, the dicing process is performed on the passivation layer **174**, the encapsulant **170** and the encapsulant **130** along the scribe lines SL. In some embodiments, since the thermal conductive pattern **162** extend beyond the scribe line SL, the dicing process is further performed on thermal conductive patterns **162**. The dicing process may be a mechanical blade sawing or a laser cutting. In some embodiments, after dicing, the sidewalls of the thermal conductive patterns **162** are exposed and substantially flush with the sidewalls of the encapsulants **120**, **170** and the passivation layer **174**. Then, a plurality of conductive terminals **176** are formed to electrically connect to the integrated circuit **110**. For example, the conductive terminals **176** are electrically connected to the integrated circuit **110** by through substrate vias (not shown) in the semiconductor substrate **112**. However, the disclosure is not limited thereto. The semiconductor package **100** may have other suitable configurations. For example, the semiconductor package **100** is formed without the conductive terminals **176**, and the semiconductor package **100** is then attached to a carrier substrate with a heat spreader thereover. In some embodiments, the formed semiconductor package **100** is a system-on-integrated-chips (SoIC) package, and the semiconductor package **100** integrates the integrated circuits **110**, **130** and the thermal conductive patterns such as the through vias **160a**, **160b** and the thermal conductive patterns **162**. However, the disclosure is not limited thereto. The semiconductor package **100** may have other suitable configurations.

(34) Referring to FIG. 1G, in some embodiments, the semiconductor package **100** is further mounted on a package substrate **180**, and a heat spreader **190** is formed to cover the semiconductor package **100**. The package substrate **180** may include a package. The package substrate **180** may be an electrical device, a printed circuit board (PCB), an interposer or other suitable redistribution line structure. In some embodiments, the package substrate **180** includes a dielectric layer **182**, a plurality of conductive patterns **184** in the dielectric layer **182** and a plurality of conductive terminals **186** disposed at a side of the dielectric layer **182** and electrically connected to the conductive patterns **184**. In some embodiments, the semiconductor package **100** is bonded to the package substrate **180** through the conductive terminals **176**, and an underfill **188** is formed between the semiconductor package **100** and the package substrate **180** to encapsulate the conductive terminals **176**. In some embodiments, the underfill **188** is further disposed on the sidewall of the encapsulant **120** and exposes the sidewalls of the thermal conductive patterns **162**. In alternative embodiments, the underfill **188** covers a portion of the sidewalls of the thermal conductive patterns **162**.

(35) In some embodiments, the heat spreader **190** is adhered to the package substrate **180** through an adhesive layer **192**. The adhesive layer **192** may be a conductive paste such as a silver paste or

an adhesive such as DAF. In some embodiments, a thermal interface material (TIM) **194** is further formed between the heat spreader **190** and the semiconductor package **100**. In some embodiments, the heat spreader **190** includes a lid, a heat sink, a water cooling device, or a combination thereof. The heat spreader **190** may surround the sidewall of the semiconductor package **100** and covers the top of the semiconductor package **100**. For example, the heat spreader **190** is immediately adjacent to the exposed sidewalls of the thermal conductive patterns **162**, and thus the heat generated by the semiconductor package **100** may be conducted to the heat spreader **190** through the thermal conductive patterns **162**. In alternative embodiments, the heat spreader **190** is in physical contact with the exposed sidewalls of the thermal conductive patterns **162**. In some embodiments, the TIM **194** is disposed between and in physical contact with the passivation layer **174** and the heat spreader **190**. Accordingly, the heat spreader **190** may be thermally coupled to the semiconductor package **100** at least through the thermal conductive patterns **162** and the TIM **194**, and thus the heat generated from the integrated circuits **110**, **130** may be dissipated efficiently. The heat spreader **190** may have other configurations. For example, the heat spreader **190** includes a first lid covering the sidewall and the top of the semiconductor package **100**, a second lid on the first lid and a heat sink on the second lid. In addition, the TIM **194** may be omitted. In such embodiments, the heat spreader **190** is in direct contact with the passivation layer **174**, the encapsulant **170**, the seed layer **140** or the integrated circuit **130**.

(36) In some embodiments, the through vias **160a** (e.g., internal through vias) and the through vias **160b** (e.g., external through vias) are arranged to surround the integrated circuits **130**, so as to thermally couple to the integrated circuits **110**, **130**. In addition, the thermal conductive patterns **162** may be formed to physically connect to the through vias **160b** and have a larger size than the through vias **160b**, and thus the heat generated by the integrated circuits **110**, **130** may be conducted through the thermal conductive patterns **162**. Thus, the heat generated by the semiconductor package **100** may be dissipated efficiently.

(37) In some embodiments, the through via **160a**, the through via **160b** and the thermal conductive pattern **162** have a height smaller than the integrated circuits **130**. However, the disclosure is not limited thereto. In alternative embodiments, as shown in FIG. 3, the through via **160a**, the through via **160b** and the thermal conductive pattern **162** have a height substantially equal to the encapsulant **170**. In such embodiments, the surfaces (e.g., top surfaces) of the through via **160a**, the through via **160b** and the thermal conductive pattern **162** are substantially coplanar with the surfaces (e.g., top surfaces) of the encapsulant **170** and the integrated circuits **130**. For example, the surfaces (e.g., top surfaces) of the through via **160a**, the through via **160b** and the thermal conductive pattern **162** are substantially coplanar with the surfaces (e.g., top surfaces) of the encapsulant **170** and the seed layer **140**.

(38) In some embodiments, adjacent through vias **160a**, **160b** are physically separated by the encapsulant **170** therebetween. However, the disclosure is not limited thereto. In some embodiments, as shown in FIG. 4A and FIG. 4B (e.g., FIG. 4A is a schematic cross-sectional view along line I-I' of FIG. 4B), adjacent two of the through vias **160a**, **160b** are physically connected by a connecting part **164**, **166**, **168** therebetween. For example, the adjacent through vias **160a** and **160b** are physically connected by the connecting part **164** therebetween, the adjacent through vias **160a** are physically connected by the connecting part **166** therebetween, and the adjacent through vias **160b** are physically connected by the connecting part **168** therebetween. The configuration, material and forming method of the connecting parts **164**, **166**, **168** may be similar to or the same as those of the thermal conductive pattern **162**. In some embodiments, the connecting parts **164**, **166**, **168** are line-shaped, respectively. However, the disclosure is not limited thereto. The connecting parts **164**, **166**, **168** may have other suitable shape. The connecting parts **164**, **166**, **168** may be formed simultaneously with the through vias **160a**, **160b** and the thermal conductive pattern **162**. For example, as shown in FIG. 4B, the through vias **160a**, **160b**, the thermal conductive pattern **162** and the connecting parts **164**, **166**, **168** are integrally formed as a net

structure. In other words, no interface exists between the connecting parts **164**, **166**, **168** and the through vias **160a**, **160b**. In some embodiments, as shown in FIG. 4A, the connecting parts **164**, **166**, **168** have a height smaller than the integrated circuits **130**. In such embodiments, the encapsulant **170** covers the connecting parts **164**, **166**, **168**, and the encapsulant **172** has a recess **172** exposing at least a portion of through vias **160a** and **160b**, the thermal conductive pattern **162** and the connecting parts **164**, **166**, **168**. For example, the recess **172** exposes a portion of the through via **160a**. In alternative embodiments, the recess **172** is omitted. In alternative embodiments, as shown in FIG. 5, the connecting parts **164**, **166**, **168** have a height substantially the same as the integrated circuits **130**.

(39) In some embodiments, at least one of the thermal conductive patterns **162** and the connecting parts **164**, **166**, **168** may be omitted. For example, as shown in FIG. 6, the through via **160a** and the through via **160b** are connected while the through vias **160a** are physically separated and the through vias **160b** are physically separated. Similarly, as shown in FIG. 7, the through vias **160a** are connected and the through vias **160b** are connected while the through via **160a** is physically separated from the through via **160b**. In above embodiments, the through vias **160a**, **160b**, the thermal conductive patterns **162** and/or the connecting parts **164**, **166**, **168** of the semiconductor package have similar arrangement and/or configuration. However, the disclosure is not limited thereto. The through vias **160a**, **160b**, the thermal conductive patterns **162** and/or the connecting parts **164**, **166**, **168** at different regions may have different arrangement and/or configuration. For example, the through vias **160a**, **160b**, the thermal conductive patterns **162** and/or the connecting parts **164**, **166**, **168** at one side of the integrated circuit **130** is different from the thermal conductive patterns **162** and/or the connecting parts **164**, **166**, **168** at another side of the integrated circuit **130**. In addition, in some embodiments, the connecting part **168** is configured to physically connect adjacent two through vias **160b**. However, the disclosure is not limited thereto. In alternative embodiments, the connecting part **168** is disposed between to physically connect adjacent two thermal conductive patterns **162**.

(40) In accordance with some embodiments of the disclosure, a semiconductor package includes a first integrated circuit, a plurality of first through vias and a plurality of fin-shaped through vias. The first through vias surround the first integrated circuit. The fin-shaped through vias are physically connected to the first through vias respectively, wherein the first through vias are disposed between the first integrated circuit and the fin-shaped through vias.

(41) In accordance with some embodiments of the disclosure, a semiconductor package includes a first integrated circuit, a plurality of first through vias, a plurality of second through vias and a first connecting part. The first through vias surround the first integrated circuit. The second through vias surround the first integrated circuit, and the first through vias are disposed between the second through vias and the first integrated circuit. The first connecting part physically connects one of the first through vias and one of the second through vias.

(42) In accordance with some embodiments of the disclosure, a semiconductor package includes a first integrated circuit, at least one second integrated circuit, a plurality of first through vias, a plurality of thermal conductive patterns and an encapsulant. The second integrated circuit is bonded to the first integrated circuit. The first through vias surround the at least one second integrated circuit over the first integrated circuit. The first through vias are disposed between the thermal conductive patterns and the at least one second integrated circuit. The encapsulant encapsulates the at least one second integrated circuit, wherein a sidewall of the thermal conductive pattern is substantially flush with a sidewall of the encapsulant.

(43) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such

equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. Semiconductor package, comprising: a first integrated circuit; a plurality of first through vias surrounding the first integrated circuit; and a plurality of fin-shaped through vias physically connected to the first through vias respectively, wherein the first through vias are disposed between the first integrated circuit and the fin-shaped through vias.
2. The semiconductor package of claim 1, wherein a width of the first through via is different from a width of the fin-shaped through via physically connected thereto.
3. He semiconductor package of claim 1 further comprising a second integrated circuit, wherein the first integrated circuit, the first through vias and the fin-shaped through vias are disposed on the second integrated circuit.
4. He semiconductor package of claim 1, wherein top surfaces of the first through vias and the fin-shaped through vias are disposed between a top surface and a bottom surface of the first integrated circuit.
5. He semiconductor package of claim 1 further comprising a second integrated circuit and an encapsulant encapsulating the second integrated circuit, wherein the first integrated circuit, the first through vias and the fin-shaped through vias are disposed on the second integrated circuit, and a sidewall of the fin-shaped through via is substantially flush with a sidewall of the encapsulant.
6. He semiconductor package of claim 1 further comprising an encapsulant covering the first through vias and the fin-shaped through vias, wherein a sidewall of the fin-shaped through via is substantially flush with a sidewall of the encapsulant.
7. The semiconductor package of claim 6, wherein a top surface of the encapsulant is coplanar with a top surface of the first integrated circuit.
8. He semiconductor package of claim 1 further comprising a seed layer, wherein the seed layer is continuously disposed beneath the first through vias and the fin-shaped through vias and on a sidewall and a surface of the first integrated circuit.
9. Semiconductor package, comprising: a first integrated circuit; a plurality of first through vias surrounding the first integrated circuit, wherein one of the first through vias has a sidewall and a first surface; a plurality of second through vias surrounding the first integrated circuit, the first through vias disposed between the second through vias and the first integrated circuit; a first connecting part, wherein the first connecting part physically connects one of the first through vias and one of the second through vias; and an encapsulant, encapsulating the first integrated circuit and the first through vias, wherein the encapsulant covers the first surface of the one of the first through vias.
10. He semiconductor package of claim 9, wherein a width of the first connecting part is different from a diameter of the one of the first through vias and a diameter of the one of the second through vias, a height between opposite surfaces of the first connecting part is equal to a height between the first surface and a second surface opposite to the first surface of the one of the first through vias.
11. The semiconductor package of claim 9, wherein a width of the first connecting part is smaller than a diameter of the one of the first through vias and a diameter of the one of the second through vias.
12. The semiconductor package of claim 9 further comprising a second connecting part, wherein the second connecting part physically connects adjacent two of the second through vias.
13. He semiconductor package of claim 12, wherein the first connecting part is connected to the second connecting part through the second through via therebetween.
14. He semiconductor package of claim 9, wherein the first connecting part is line-shaped.

15. The semiconductor package of claim 9, wherein no interface exists between the first connecting part and the one of the first through vias and between the first connecting part and the one of the second through vias.
16. Semiconductor package, comprising: a first integrated circuit; at least one second integrated circuit bonded to the first integrated circuit; a plurality of first through vias surrounding the at least one second integrated circuit over the first integrated circuit; a plurality of thermal conductive patterns, the first through vias disposed between the thermal conductive patterns and the at least one second integrated circuit; and an encapsulant encapsulating the at least one second integrated circuit, wherein a sidewall of the thermal conductive pattern is substantially flush with a sidewall of the encapsulant.
17. He semiconductor package of claim 16 further comprising a heat spreader, wherein the heat spreader surrounds the first integrated circuit, the at least one second integrated circuit, the first through vias and the thermal conductive patterns.
18. He semiconductor package of claim 16 further comprising a heat spreader, wherein the heat spreader covers the at least one second integrated circuit, the first through vias and the thermal conductive patterns.
19. He semiconductor package of claim 16, wherein the encapsulant exposes at least one of the first through vias and the thermal conductive patterns.
20. The semiconductor package of claim 16, wherein a surface of the encapsulant is substantially coplanar with a surface of the at least one second integrated circuit.
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